

Methodology - Art Haxholli

Research and verification process for the ferroelectric HfO₂ memory ontology

This ontology sketch was developed in a two phase style: an AI-assisted literature survey, followed by a human-directed verification process. The first phase used Claude to search the web, retrieve and read technical literature, and draft a structured research base covering the topics required by the task. I supplied the model with the task definition, conditions and scope. The output was an unchecked reference document that simply served as a first draft for structure and orientation.

Before verifying the draft, I began inspecting a sample of the sources that were used by the model to become familiar with the topic's fundamental ideas, and to determine which of those sources were most aligned with the task. After that, I began inspecting the draft. The model was explicitly asked to extract production flows from the sources and cite what was used. The information appeared to be properly cited and not hallucinated. I reverse engineered one of the flows that was given in the document, and also took a sample of citations to see if there were any hallucinated facts. There were no such cases. I also ran different sessions to see whether different agents would find issues, creating a feedback loop of corrections. In the end I determined that the facts were extracted properly, but in a lot of cases the models had cited abstracts or snippets of papers instead of accessing the paper's contents, which led to the information being interpreted incorrectly. This led to me gathering a subset of the papers that covered most of the document, dropping the other ones, and then incrementally feeding the papers to the model. This allowed for a more structured, manageable and richer approach to information extraction. The paper count was reduced from 28 to 10.

The resulting ontology specification is grounded exclusively in those ten sources, each read in full or near-full, with per-section locators recorded for every assertion. Uncertainties have been recorded, whether because of sources, or the research limitations. Modelling decisions that no source states are inferred. The ontology sketch itself, and the final judgement on which assertions to keep, are my own work. The model's role was information retrieval, drafting, structuring, and once primary sources were supplied, auditing its own prior output against them.